Xintong Zhang

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RESEARCH INTEREST

Embodied AI, World Models, Robot Learning, Vision-and-Language Navigation

EDUCATION

Duke Kunshan University (DKU) & Duke University Dual Degree Aug 2023-May 2027
Major: Applied Mathematics and Computational Sciences / Computer Science Kunshan, China/Durham, USA

- GPA: 3.976/4.0
- Class Ranking: Top 5%
- Dean's List with Distinction (2023 Fall, 2024 Spring, 2024 Fall, 2025 Spring, 2025 Fall), Dean's List (2026 Spring)
- Skills: Python, Java, C++, LaTex, Gurobi, Unreal Engine, Airsim, NumPy, OpenCV, ROS (Robot Operating System), Linux, Git, Robot Hardware (Motive, Crazyfly UAV, Tello UAV, SCOUT UGV), Blender

PUBLICATIONS AND MANUSCRIPTS

- Kunhui Wang*, **Xintong Zhang***, Junyu Gao, Changsheng Xu, "ForeFly: A Dual-Horizon World Action Model for Aerial Vision-Language Navigation," *The Fifteenth International Conference on Learning Representations (ICLR 2027)*. Under Review. Paper
- Kunhui Wang, **Xintong Zhang**, Yufan Hu, Yang Yang, and Junyu Gao, "MoARFly: Learning Motion-Aware Action Refinement for Aerial Vision-Language Navigation," *IEEE Transactions on Multimedia (TMM)*. Under Review
- Junfeng Chen, Yuxiao Zhu, An Zhuo, **Xintong Zhang**, Shuo Zhang, Guanghui Wen, Xiwang Dong, Meng Guo, Zhongkui Li, "Melding LLM and temporal logic for reliable human-swarm collaboration in complex environments," *npj Robotics*. Under Review. Paper Project Page Video
- **Xintong Zhang***, Junfeng Chen*, Yuxiao Zhu, Bing Luo and Meng Guo, "CoCoPlan: Adaptive Coordination and Communication for Multi-robot Systems under Continual Temporal Tasks," *Robotics and Automation Letters (RA-L)*. Accepted. Paper Project Page Video
- Yuxiao Zhu*, Junfeng Chen*, **Xintong Zhang**, Meng Guo and Zhongkui Li, "DEXTER-LLM: Dynamic and Explainable Coordination of Multi-Robot Systems in Unknown Environments via Large Language Models," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)*. Accepted. Paper Project Page Video
- Junfeng Chen, Yuxiao Zhu, **Xintong Zhang**, Bing Luo and Meng Guo, "SLEI3D: Simultaneous Large-scale 3D Exploration Inspection and Interaction via Heterogeneous Fleets under Limited Communication," *IEEE Transactions on Automation Science and Engineering (TASE)*. Accepted. Paper Project Page

RESEARCH EXPERIENCE

Institute of Automation, Chinese Academy of Science — Undergraduate Researcher
Supervisors: Prof. Junyu Gao, Prof. Changsheng Xu

1. **ForeFly: A Dual-Horizon World Action Model for Aerial Vision-Language Navigation (ICLR: Under Review)**
   May 2026-Now
   Co-First Author Beijing & Kunshan, China
   - Designed the Foresight-Guided Action Refinement (FGAR) to incorporate predicted proximal and route-critical future-world representations into action generation, improving the use of future information for both local motion continuity and long-horizon navigation.
   - Developed image reconstruction visualizations using Stable Diffusion v1-5 to analyze predicted future representations.
   - Conducted extensive experiments and ablation studies and performed both quantitative and qualitative evaluations.
   - The method achieved state-of-the-art performance (41.75% success rate) on the widely used TravelUAV benchmark.
2. **MoARFly: Learning Motion-Aware Action Refinement for Aerial Vision-Language Navigation (TMM: Under Review)**
   Apr 2026-Aug 2026
   Second Author Beijing, China
   - Contributed to the design of the Dual-Query Transformer architecture, integrating visual, motion, action-query, and language representations.
   - Conducted experimental analysis and contributed to manuscript writing.

Peking University — Undergraduate Researcher
Supervisor: Prof. Meng Guo

1. **LLM-Driven Planning and Coordination for Heterogeneous Multi-Robot Systems** Jan 2025-Mar 2026
   Publications: DEXTER-LLM, IROS; Melding LLM and temporal logic (Under Review) Beijing & Kunshan, China
   - Developed LLM-assisted task-planning frameworks for heterogeneous multi-robot systems in unknown environments, integrating LLM-based high-level reasoning with model-based optimization for explainable and reliable coordination.
   - Designed a branch-and-bound (BnB)-based subtask assignment and scheduling algorithm and developed structured prompt templates for LLM-based multi-stage reasoning and subtask generation.

- Conducted physical hardware experiments by deploying planning algorithms on physical robot systems involving 2 UAVs, 4UGVs and 2 quadruped robots, enabling real-world validation of the proposed planning framework.

2. **CoCoPlan: Adaptive Coordination and Communication for Multi-robot Systems under Continual Temporal Tasks (RAL: Accepted)** Jan 2025-Aug 2025
   Leading Author Beijing & Kunshan, China
   - Led the design of a framework that co-optimizes task planning and intermittent communication based on a BnB architecture and proposed an adaptive objective function that balances task efficiency against communication latency.
   - Developed a communication planning approach combining tree-search methods and nonlinear iterative optimization, supporting 2D and 3D environments, regular/irregular obstacles, and scenarios with up to 100 agents in 15 seconds.
   - Conducted hardware experiments in 138*114*3 m^3 long narrow corridor environment with 4 UGVs, and 16*14*26 m^3 disaster environment with 2 UAVs and 4UGVs.
3. **SLEI3D: Simultaneous Large-scale 3D Exploration Inspection and Interaction via Heterogeneous Fleets under Limited Communication (TASE: Accepted)** May 2024-Dec 2024
   Third Author Beijing & Kunshan, China
   - Designed a multi-layer, multi-rate coordination framework for heterogeneous UAV/UGV teams, jointly optimizing exploration, inspection, and inter-robot communication.
   - Implemented low-level robot motion planning in C++ and high-level planning/optimization algorithms in Python, including TSP, multi-vehicle routing, and genetic algorithms.
   - Conducted large-scale 3D simulations with up to 48 robots in a 384,000 m^3 environment in RViz, and performed hardware experiments with 4 Crazyflie UAVs, 2 Tello UAVs, and 1 UGV.

COURSE PROJECTS

Duke University — Research Independent Study
Development of Multi-Agent Simulation Environment Powered by Large Language Models Sep 2025–Dec 2025
Supervisor: Prof. Boyuan Chen Durham, NC, USA

- Developed a LLM powered framework that allows users to specify operational areas, heterogeneous agents, tasks, constraints, and task-allocation strategies for customized multi-agent collaboration problems.
- Integrated LLMs for strategy validation, semantic task decomposition, and capability-aware subtask assignment, connecting high-level natural-language problem specifications with executable agent behaviors in a structured grid world.
- Implemented multi-agent task execution and quantitative strategy evaluation and evaluated the framework across diverse scenarios including chemical-plant emergency response, medical operations, and office coordination.

ACADEMIC ACTIVITIES & SERVICES

Reviewer: RAL 2026, **Conference Participation:** IROS 2025, ICRA 2026
Teaching Assistant: DKU COMPSCI101: Introduction into Computer Science (2024 Fall)

HONORS & AWARDS

Dean's List (2026 Spring) – DKU May 2026
Dean's List with Distinction (2023 Fall, 2024 Spring, 2024 Fall, 2025 Spring, 2025 Fall) – DKU 2023-2025
National Encouragement Scholarship - Jiangsu Provincial Department of Education Dec 2025
Student Experiential Learning Fellow (SELF) – DKU Apr 2025
Signature Fund (SF) – DKU Apr 2025
National-Level College Student Innovation and Entrepreneurship Project - Organizing Committee of the Jiangsu Provincial College Student Innovation and Entrepreneurship Outstanding Achievement Exchange and Exhibition Apr 2024